DJ Desk Architecture and Product Decisions

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<!-- Source: docs/adr/0001-application-backbone.md -->

ADR 0001: Application Backbone

Status

Accepted.

Date

2026-04-20

Context

DJ Desk needs to move quickly from raw track analysis into an interactive set-planning workspace. The first product risk is the domain model: keys, modes, BPM, chord progressions, confidence states, and set planning workflows. Rendering performance is important, but it is not the bottleneck yet.

The application starts local-first, but it should be deployable later without changing the core architecture.

Decision

Use an npm workspace monorepo:

- `apps/client`: Vue 3 + Vite application.
- `apps/server`: Fastify API running TypeScript directly on Node 24 LTS.
- `packages/domain`: shared TypeScript package for musical domain types and helpers.
- `docs/adr`: architecture decisions.

Use Node 24 LTS as the runtime baseline and npm as the only package manager. Do not use Yarn Plug'n'Play or pnpm.

Use Vue 3 stable first. Do not base the application on Vue Vapor Mode yet, because Vapor is still beta-only in Vue 3.6. Keep the client in Composition API and `<script setup>` so Vapor can be evaluated later for isolated, performance-sensitive surfaces.

Use TypeScript 6 as the stable compiler. Keep TypeScript 7 native preview optional through `npm run typecheck:native`; it is not a required build step.

Use Oxlint and Oxfmt as the primary linting and formatting tools. Avoid ESLint/Prettier until a concrete rule gap appears.

Use a build boundary for `packages/domain`. Node can run application `.ts` files directly,

but built-in type stripping refuses TypeScript files under `node_modules`. The shared
package therefore emits JavaScript and declarations to `dist`.

Prepare the backend for SQLite persistence, but keep the initial API in memory while the
domain model and screens are still forming.

Consequences

- The development stack stays small and fast.
- The server can run `.ts` directly with Node's stable type stripping.
- Type checking remains explicit via `tsc`; Node does not type-check runtime TypeScript.
- Shared domain logic is isolated from framework code and can be tested independently.
- The app remains easy to deploy later: build the domain package, build the client, and
  run the Fastify server on Node 24 LTS.
- Vapor Mode can be tested later without forcing the whole client onto a beta runtime.

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<!-- Source: docs/adr/0002-modal-circle-workspace.md -->

ADR 0002: Modal Circle Workspace

Status

Accepted.

Date

2026-04-20

Context

The main DJ Desk surface is a production workspace for manually arranging tracks into a
set. The interface must help the operator reason about harmonic proximity quickly, while
still leaving room for manual judgement.

Most tracks can be placed by tonic and common mode, but Brazilian and adjacent repertoire
often contains modal color, variable scale degrees, mixture, or unusual modulations.
Those cases need to be visible without forcing a full music-theory analysis into the
main map.

The future dataset is expected to contain hundreds of tracks, so the first screen must
support dense browsing, quick focus changes, and a draft set chain.

Decision

Use a twelve-section circle-of-fifths workspace as the primary visual model. The first
solmization labels are fixed to Do, Re, Mi, Fa, Sol, La, Si, with enharmonic spellings

shown together for sharp/flat pitch classes.

Track placement is derived from `TrackKey` and represented as `ModalPlacement`:

- `home`: major and natural minor tracks remain in the tonic section.
- `pure-modal`: pure Dorian, Lydian, Phrygian, and Mixolydian tracks are placed in the section representing their diatonic collection.
- `modal-mixture`: tracks with a variable degree are placed between the home tonic and the pure modal target section.

Use `variable-degree` as the data variant and `modal mixture` as the UI lane language.
This avoids calling tracks "dirty" while still capturing the practical mixing concern.

Add two independent free-text fields to tracks:

- `harmonyNotes`: special warnings or instructions about harmony, modulation, exotic modes, or other analysis details that require attention before mixing.
- `comment`: ordinary operator notes that do not need warning treatment.

When `harmonyNotes` is present, show a compact warning indicator on the track dot, in the sector browser, and inside the draft set chain. Show the full text only in the focused track panel.

Keep chord progressions as a simple list for now. The future raw dataset may contain a large amount of chord data, so the detailed chord table will be designed after import format and density are known.

Consequences

- The map stays fast to scan even when tracks have nuanced modal behavior.
- A track can be visible near both its home tonic and its modal collection through the `modal-mixture` lane and nearby-section filtering.
- Special harmony warnings are not lost inside ordinary comments.
- The model remains small enough for the spike while leaving room for richer analysis data, chord tables, solmization preferences, and import workflows.

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<!-- Source: docs/adr/0003-sqlite-persistence-and-migrations.md -->

ADR 0003: SQLite Persistence and Migrations

Status

Accepted.

Date

2026-04-20

Context

DJ Desk starts as a local-first application, but it should not keep its core dat
a only in
memory. Track analysis, harmony notes, comments, chord lists, tags, and draft se
t order
need a durable structure before raw import work begins.

The app is still a spike, so the persistence layer should stay small and easy to
 replace
if deployment requirements later push the project toward Postgres or another dat
abase.

Decision

Use SQLite as the first durable store. Store the default database at `data/djdes
k.sqlite`
and allow overrides through `DATABASE_PATH`.

Use plain SQL migration files under `apps/server/migrations`. The Fastify server
 applies
pending migrations on startup, and `npm run db:migrate` runs the same migration
path
manually.

Keep migrations structural only. Seed data comes from the existing TypeScript sa
mple
tracks and is inserted only when the database is empty. This keeps production da
ta and
local bootstrap data separate.

Create normalized tables for:

- tracks and their key, BPM, confidence, harmony note, and comment metadata.
- ordered chord symbols per track.
- tags per track.
- draft sets and ordered draft set tracks.

Consequences

- Local development has a durable database without adding an external service.
- The API can move from in-memory sample data to real persistence without changi
ng the
  client contract.
- Raw imports can target normalized tables instead of pushing large JSON blobs t
hrough
  the app.
- Future migration to another SQL database remains possible because schema chang
es are
  explicit and repository access is isolated on the server.

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<!-- Source: docs/adr/0004-modal-subsections-and-transition-rules.md -->

ADR 0004: Modal Subsections and Transition Rules

Status

Accepted.

Date

2026-04-20

Context

The imported catalog is dense enough that dots on the modal circle are not a usable primary control. DJ Desk needs larger tap targets and a clearer model for pure modal tracks and modal-mixture tracks around each circle boundary.

The operator also needs the `Add` warning and wheel coloring to follow a small, explicit transition graph instead of a broad "nearby key" heuristic.

Decision

Keep the twelve natural-minor circle sections as the main frame.

Within each section, divide the annular area into:

- a central home subsection for natural minor, major via relative minor, and raised-leading-tone home tracks.
- two side pure-modal subsections near the neighboring boundaries.
- a boundary subsection between each pair of circle sections for modal-mixture tracks.

Pure modal tracks are displayed in the section of the natural-minor collection they use, not in the section of their tonic. For example:

```text
La natural minor
Mi pure Phrygian
La Dorian mixture + Mi Phrygian mixture
La pure Dorian
Mi natural minor
```

Use these directed transition rules for the draft-set `Add` warning:

- home can move to the same or neighboring home section.
- pure modal can move to pure modal with the same mode and same or neighboring home tonic.
- home can move to pure modal when the home section is the pure modal track's home section or target collection section.
- pure modal can move to home when the home section is the pure modal track's home section or target collection section.
- home can move to modal mixture when the home section belongs to that modal-mixture boundary.
- modal mixture can move to home when the home section belongs to that modal-mixture boundary.
- modal mixture can move to modal mixture only on the same boundary.
- modal mixture can move to pure modal only for the same home tonic and same mode.

The wheel compatibility coloring stays symmetric for browsing: a zone is shown as compatible with the last draft track when at least one track in the zone can move in either direction with that draft endpoint.

Consequences

- Pure modes are easier to understand visually because they sit beside the natural-minor collection they actually use.
- Modal-mixture tracks from neighboring tonics naturally share one boundary target.
- The add button remains directional, while the wheel stays useful for browsing

both possible directions around the current draft endpoint.
- The rules are intentionally conservative and can be revised after listening tests without changing the storage model.

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<!-- Source: docs/adr/0005-default-key-display-solmization.md -->

ADR 0005: Default Key Display Solmization

Status

Accepted.

Date

2026-04-20

Context

DJ Desk stores analyzed keys as normalized pitch classes so imports, transition rules, and database queries stay compact and stable. That internal representation should not dictate how the interface displays keys.

The working UI should use the operator's default solmization vocabulary instead of raw letter names. Letter-key display may return later as an explicit display preference, but it should not leak into the default interface.

Decision

Keep `TrackKey.tonic` and database `tonic` values in normalized pitch-class form.

Use domain display formatters for user-facing key labels:

- C -> Do
- D -> Re
- E -> Mi
- F -> Fa
- G -> Sol
- A -> La
- B -> Si

For altered pitch classes, show Unicode accidentals:

- sharp: `♯`
- flat: `♭`

When an imported key is stored in the sharp-normalized internal form but has a common enharmonic flat display, show both solmization labels in the UI. The canonical spelling is shown first and the alternate spelling second.

Circle section labels keep the minor-section suffix, for example `La m` and `Fa♯ m / Sol♭ m`. Section labels use the natural-minor circle convention: `Do♯ m` remains primary for C-sharp minor, while `Mi♭ m` and `Si♭ m` are primary for the D-sharp/E-flat and A-sharp/B-flat sections.

Full track key labels combine the solmization tonic with the mode label, for example `Re Dorian`. For full key labels, choose the canonical tonic spelling by the key's mode spelling cost: prefer the spelling that avoids excessive accidentals in the full seven-note mode. For example:

- C-sharp natural minor -> `Do♯ / Re♭ Natural minor`
- C-sharp major -> `Re♭ / Do♯ Major`
- A-sharp natural minor -> `Si♭ / La♯ Natural minor`

Consequences

- The UI no longer exposes raw `A`, `B`, `C`, `D`, or `E` key names by default.
- Enharmonic pairs no longer always put the sharp spelling first.
- Importers and transition logic do not need storage changes.
- A future notation switcher can be implemented as another formatter without changing the database schema.

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<!-- Source: docs/adr/0006-dense-set-planning-interactions.md -->

ADR 0006: Dense Set-Planning Interactions

Status

Accepted.

Date

2026-04-20

Context

After importing real catalog data, the modal circle is no longer a small demo map. Each section can contain many tracks, some tracks have unknown keys, and a draft set can become harmonically invalid after removing or moving one item.

The operator needs the interface to make these states visible without blocking manual judgement. DJ Desk should warn about risky transitions, but it should not prevent intentional non-harmonic choices.

The first implementation remains local and client-side. Draft set editing is mocked in the Vue state until persisted multi-set editing is implemented.

Decision

Keep track browsing centered on sections, subsections, and boundaries, but make the selected map area explicit:

- selected parent sections, natural subsections, pure-modal subsections, and boundary subsections receive a visible selected contour.
- the center readout describes the selected map area, not the currently focused track.
- boundary readouts use compact primary section labels so long enharmonic strings do not overflow the center of the circle.

Treat unknown-key tracks as a separate shelf outside the circle:

- the unknown-key control is placed at the bottom-right of the wheel panel.
- unknown-key tracks remain hidden from harmonic sections until a key is known.
- selecting the shelf opens only unplaced tracks in the browser.

Use user-facing music language in the browser:

- show `natural tracks` instead of the internal `home tracks` label.
- show `natural`, `pure modal`, `modal mixture`, and `unknown key` as lane labels on track cards.

Keep the `Add` action permissive but stable:

- a red `Add` button means the selected track is already present in any draft set, the transition is non-harmonic, or both.
- the visible reason is shown next to the button.
- the action area reserves a fixed layout slot so the button does not jump when the reason appears or disappears.
- repeated tracks are allowed because a DJ set may intentionally reuse a track.

Expose draft set editing directly in the chain:

- tracks can be removed from any position.
- tracks can be moved with up/down controls.
- tracks can also be reordered with native drag-and-drop.
- after any edit, every adjacent transition is re-evaluated.
- risky adjacent pairs are shown as red `Non-harmonic transition` dividers between the two tracks, because non-harmonicity is a relationship between neighboring set positions rather than an attribute of either track alone.

Add a `Sort harmonically` control for the active draft set:

- the sorter is a deterministic client-side helper, not a persisted planner.
- it tries each track as a possible start and greedily chooses the next best track by the current transition rules.
- compatible transitions score highest; BPM distance and circle distance are only tie-breakers.
- if no fully compatible sequence exists, the best available order is applied and remaining risky joins stay visibly marked.

Add BPM compatibility to the track browser:

- browser tracks are sorted against the last track in the active draft set.
- the primary browser sort is a soft BPM compatibility score, followed by harmonic compatibility as a tie-breaker.
- the BPM score compares the real BPM values directly; half-time and double-time equivalence are intentionally not considered because this workflow does not mix that way.
- the score is displayed as a number and clipped progress meter on each track card.
- the progress meter uses one fixed red-yellow-green scale and clips the unfilled right side, so short bars do not rescale the full gradient.

Keep verification confidence close to the values it qualifies:

- track cards do not show large standalone `Key unverified` or `BPM unverified` badges.
- key and BPM confidence labels are rendered inline next to the corresponding key or BPM value.
- the focus panel uses the same inline confidence pattern.

Keep the future focused-view idea documented but out of scope for this change: a later map mode may show only the active sector and neighboring sectors instead of the full circle, improving tap targets on small screens.

Consequences

- Dense sections are easier to operate because the selected browsing target is visible independently from the last draft track.
- Unknown-key tracks are no longer presented as if they belonged to the harmonic map.
- The operator can intentionally create, inspect, and repair non-harmonic joins.

- Non-harmonic warnings are easier to interpret because they are attached to transitions, not track cards.
- BPM proximity becomes visible during next-track browsing without imposing a hard compatibility cutoff.
- Track cards are less visually noisy while still preserving unverified key and BPM information.
- The draft set chain becomes editable without introducing a persistence layer change yet.
- Harmonic sorting is useful as a fast repair tool but remains explainable and non-authoritative.
- Future work should persist draft set changes, add undo/redo, and consider a focused sector view for mobile and laptop layouts.

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<!-- Source: docs/harmonic-transition-rules.md -->

Harmonic Transition Rules

DJ Desk treats every analyzed key as a transition profile. The profile is used for the draft-set `Add` warning and for compatibility coloring on the circle.

Profiles

Home

Regular major, natural minor, and raised-leading-tone minor tracks are `home` tracks. A major track is placed in the section of its relative minor.

```text
C major -> La m section
A natural minor -> La m section
G natural minor with raised leading tone -> Sol m section
```

Pure Modal

Pure modal tracks keep their modal home tonic in the profile, but the UI places them in the section of the natural-minor pitch collection they use.

```text
A Dorian pure:    home La m, displayed in Mi m collection
E Phrygian pure:  home Mi m, displayed in La m collection
C# Phrygian pure: home Do# m, displayed in Fa# m collection
```

Dorian and Lydian point clockwise from their home tonic to the collection section. Phrygian and Mixolydian point counter-clockwise. On the wheel this means the pure modal side zones around a boundary read as:

```text
La natural minor
E pure Phrygian
La Dorian mixture + E Phrygian mixture
La pure Dorian
Mi natural minor
```

Modal Mixture

Modal-mixture tracks live on the boundary between their home section and modal target section. Neighboring modal-mixture tracks from both sides of the same

boundary share one boundary zone.

```text
A Dorian modal mixture:    La m / Mi m boundary
E Phrygian modal mixture:  Mi m / La m boundary
C# Phrygian modal mixture: Do# m / Fa# m boundary
```

Directed Add Rules

The draft-set `Add` warning is directional: it checks whether the selected track can follow the last draft track.

- `home -> home` is valid when the sections are the same or adjacent.
- `home -> pure modal` is valid when the home section matches either the pure modal track's home section or target collection section.
- `pure modal -> home` is valid when the next home section matches either the pure modal track's home section or target collection section.
- `pure modal -> pure modal` is valid when both tracks use the same modal mode and their home sections are the same or adjacent.
- `home -> modal mixture` is valid when the home section belongs to that modal-mixture boundary.
- `modal mixture -> home` is valid when the home section belongs to that modal-mixture boundary.
- `modal mixture -> modal mixture` is valid only when both tracks use the same boundary.
- `modal mixture -> pure modal` is valid only into the same tonic and same mode.

No other cross-lane transition is considered valid in the current spike.

Reference Cases

```text
D natural minor -> A natural minor        valid
D natural minor -> A Dorian pure          warning
D Dorian pure -> A Dorian pure            valid

A natural minor -> E natural minor        valid
A natural minor -> A Dorian pure          valid
A natural minor -> A Phrygian pure        valid
A natural minor -> E Dorian pure          warning
A natural minor -> E Phrygian pure        valid
A natural minor -> E Dorian mixture       warning

A natural minor -> A Dorian mixture       valid
A natural minor -> E Phrygian mixture     valid
A Dorian mixture -> E natural minor       valid
A Dorian mixture -> A natural minor       valid
A Dorian mixture -> A Dorian pure         valid
A Dorian mixture -> E Dorian pure         warning
A Dorian mixture -> E Phrygian mixture    valid

F# natural minor -> C# Phrygian mixture   valid
D natural minor -> A Dorian mixture       warning
```

Draft Reference Coloring

Circle coloring uses the last track in the draft set as the reference track, because the map should answer the planning question "what can I pick next?". Selecting a row or a circle zone changes the focus panel and browser content, bu
t

does not change the compatibility colors.

A zone is compatible with the last draft track when at least one track in that zone can either follow the draft track or precede it. This symmetric rule keeps the map useful when browsing around the current draft endpoint.

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<!-- Source: docs/ideas.md -->

Product Ideas

Focused Sector View

Add a second circle-display mode for detailed planning on small screens and dens
e
key areas.

Instead of showing the whole circle, the view would show the active sector, its two neighboring sectors, and the two boundaries between them. This should make the areas that matter for the next transition much larger and easier to tap, especially on phones.

Open questions:

- Whether this mode should open on tap, via a segmented control, or both.
- Whether boundary modal-mixture zones should appear as full columns between sectors in this focused view.
- Whether the draft set should automatically move the focused view to the last added track's section.